

End-to-End Quantum-Fuzzy Logical Petri Nets for Safety-Critical Event Supervision: Cross-Modal Corroboration, Robust Decision, and Evidence-Based Validation

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2026

Abstract

Safety-critical cyber-physical systems must combine discrete event logic, uncertain sensor evidence, state evolution, and explicit assurance mechanisms without conflating these semantic layers. This paper develops and evaluates an end-to-end Quantum-Fuzzy Logical Petri Net (QFLPN) supervision pipeline for a controlled safety-critical event-detection problem. The application model contains four noisy sensor channels, a four-mode discrete supervisory state, fuzzy membership functions, transition activation, a two-qubit operator encoding, and a final emergency decision. The quantum layer is implemented as an exact classical state-vector simulation; no physical quantum hardware claim is made. The central experimental question is whether requiring corroboration between two independent sensor groups improves robustness when a single sensor is progressively biased. The benchmark contains 300 deterministic-seed episodes of 240 samples each, with three fault families, 180 training episodes and 120 held-out test episodes. Thresholds are selected only on the training set. On the clean test set, the QFLPN decision reaches an F1 score of 0.8499, precision 0.8751, recall 0.8262, and false-positive rate 0.0247. The crisp guard obtains F1=0.9084, while fuzzy fusion obtains F1=0.8421. Under a temperature-sensor bias of 1.2 normalized units applied only to normal test samples, the crisp guard false-alarm rate rises to 0.9933, fuzzy fusion to 0.1443, whereas QFLPN remains at 0.0859. At the strongest tested bias of 1.8, QFLPN retains F1=0.7390 while the crisp guard falls to 0.2760 and fuzzy fusion to 0.6084. The evidence therefore supports a conditional claim: cross-modal QFLPN corroboration provides a substantially more stable safety-decision boundary under single-sensor corruption in the tested synthetic workload. The result is an application-level validation, not a proof of universal superiority, hardware quantum advantage, or industrial certification.

Keywords: Quantum-Fuzzy Logical Petri Nets; safety-critical systems; cyber-physical systems; fuzzy supervision; quantum state encoding; Petri nets; runtime assurance; sensor corruption; robustness; reproducible benchmarking.

1 Introduction

Cyber-physical systems (CPS) integrate sensing, computation, communication, actuation, and physical processes whose interactions may have safety, timing, reliability, and security consequences. NIST emphasizes that trustworthy CPS engineering requires explicit treatment of functional behaviour, timing, composition, boundaries, lifecycle, and assurance rather than isolated algorithmic performance [5]. This requirement becomes particularly demanding when sensor information is uncertain, event logic is discrete, and the computational state is represented through a quantum or quantum-inspired operator layer.

Petri nets are a natural representation for concurrent and discrete-event structure because places, transitions, markings, and arcs provide an explicit execution semantics [1]. Fuzzy extensions can represent graded evidence when binary transition guards are too rigid [2]. Quantum information processing introduces a different state semantics based on Hilbert spaces, amplitudes, density operators, and measurement probabilities [3]. The scientific difficulty is therefore not to attach quantum terminology to a Petri net, but to define a typed interface between these layers and then demonstrate that the resulting system has a measurable application-level benefit.

The present paper addresses the application-level question. The objective is an end-to-end supervisory pipeline in which sensor observations generate fuzzy evidence, fuzzy evidence activates discrete supervisory transitions, the activated information is encoded into a quantum state, and the resulting measurement probability contributes to a safety decision. The paper deliberately uses a controlled synthetic CPS workload so that every signal, fault onset, perturbation, random seed, threshold, and evaluation metric is reproducible. This choice provides experimental control while preventing an unsupported claim of industrial validation.

The scientific question is narrower than a general claim about quantum computing for safety. The question is whether a QFLPN decision rule that requires corroboration between two sensor groups can maintain a useful safety-decision boundary when one sensor is progressively corrupted. The comparison is therefore against a crisp event guard and a conventional fuzzy fusion rule under the same data, training protocol, thresholds, and test episodes.

The contribution is application-level and evidence-oriented. It combines a discrete supervisory state machine, fuzzy activation, a quantum probability interface, fault-mode simulation, controlled sensor corruption, held-out evaluation, and explicit claim boundaries. The result is intended to be useful as a blueprint for future CPS validation rather than as a certification argument.

2 Scientific Position and Related Work

CPS assurance is inherently interdisciplinary. The NIST CPS framework identifies trustworthiness, timing, data, composition, boundaries, and lifecycle as interacting engineering concerns [5]. Earlier CPS research similarly emphasizes the tight coupling between computational algorithms and physical dynamics [6]. Safety analysis therefore benefits from architectures in which assumptions and evidence remain traceable.

Petri-net research provides a mature basis for discrete-event modelling, concurrency, reachability, and synchronization [1]. Fuzzy logic provides a mathematically explicit representation of graded membership and uncertain concepts [2]. Fuzzy Petri-net research has extended this combination toward rule-based inference and uncertain event interpretation. Recent work continues to use fuzzy Petri structures for critical-infrastructure protection and vulnerability assessment [11].

Runtime verification provides a complementary perspective: instead of proving every possible implementation behaviour statically, a monitor evaluates observed execution traces against explicit properties. Recent CPS work combines temporal logic with runtime monitoring for anomaly prediction and security assurance [12]. Other recent work combines formal verification and runtime monitoring to bridge the gap between verified models and executable CPS components [13]. These approaches motivate the present emphasis on explicit state, decision thresholds, and measurable assurance metrics.

Quantum information theory provides the mathematical foundation for state vectors, unitary operators, tensor products, and measurement probabilities [3]. Quantum open-system theory extends the state description to density operators and completely positive trace-preserving maps [4]. The present application does not require open-system dynamics because its scientific purpose is different: it isolates the decision-level value of an operatorial quantum encoding under sensor uncertainty. Consequently, no claim about decoherence resistance or physical quantum hardware

is made.

The relationship to recent Petri-net CPS software is also instructive. RIPN integrates interpreted Petri nets with teleo-reactive programs for CPS implementation [14]. That line of work demonstrates the continuing relevance of executable Petri-based architectures. The present study differs by making fuzzy sensor evidence and a quantum-state encoding explicit in the supervisory path and by measuring robustness to a controlled sensor corruption rather than focusing on software architecture alone.

3 Application Problem

The application is a synthetic safety-critical supervisory controller for a process whose physical state is observed through temperature, vibration, pressure, and current sensors. Three fault families are considered. The first represents a thermal-pressure disturbance, the second a mechanical disturbance, and the third a pressure-dominated disturbance. Each fault is intentionally multi-modal: a physically meaningful event affects more than one sensing modality.

The supervisory state contains four logical modes: Normal, Warning, Degraded, and Emergency. The invariant is that exactly one supervisory mode is active at any sampling instant. The state transition logic is intentionally discrete, while the sensor evidence is continuous-valued and noisy. The resulting problem is therefore hybrid at the modelling level even though the benchmark uses discrete sampling.

The benchmark does not attempt to reproduce a particular industrial plant. It is a controlled numerical experiment designed to isolate one scientific mechanism: cross-modal corroboration. A normal state may contain noise and benign excursions, whereas a fault state contains a deterministic severity trajectory contaminated by stochastic measurement noise. A sensor corruption experiment subsequently adds a systematic positive bias to temperature measurements while leaving the ground-truth physical state unchanged.

4 End-to-End QFLPN Model

Let the four sensor observations at time k be

$$s_k = (T_k, V_k, P_k, I_k), \quad (1)$$

where T denotes temperature, V vibration, P pressure and I current. Each sensor is mapped to a fuzzy membership degree through a logistic function

$$\mu_j(s) = \frac{1}{1 + \exp(-(s - b_j)/a_j)}. \quad (2)$$

The thermal-current evidence group and vibration-pressure evidence group are defined as

$$\lambda_A = 1 - (1 - \mu_T)(1 - 0.55\mu_I), \quad (3)$$

and

$$\lambda_B = 1 - (1 - \mu_V)(1 - 0.65\mu_P). \quad (4)$$

Both values lie in $[0, 1]$. The construction intentionally requires corroboration across two different modalities rather than allowing one sensor to dominate the final decision.

The supervisory Petri layer is represented by places

$$P = \{P_N, P_W, P_D, P_E\}, \quad (5)$$

corresponding to Normal, Warning, Degraded and Emergency. A marking contains exactly one supervisory token. Transition guards depend on the current fuzzy evidence and on the persistence of the preceding mode. This gives the application a discrete execution contract without identifying fuzzy membership with a quantum amplitude or measurement probability.

The quantum layer maps the two fuzzy group activations into a two-qubit state. For each group,

$$\theta_i = 2 \arcsin \sqrt{\lambda_i}, \quad i \in \{A, B\}, \quad (6)$$

and

$$|\psi_k\rangle = (R_y(\theta_A) \otimes R_y(\theta_B)) |00\rangle. \quad (7)$$

The probability of the corroborated basis state $|11\rangle$ is

$$p_{11}(k) = |\langle 11 | \psi_k \rangle|^2 = \lambda_A(k) \lambda_B(k). \quad (8)$$

This distinction is important. λ_A and λ_B are fuzzy memberships, while p_{11} is a quantum measurement probability. The equality $p_{11} = \lambda_A \lambda_B$ is a consequence of the declared encoding and product-state construction; it is not a statement that fuzzy membership is intrinsically a quantum probability.

The final QFLPN alarm score is

$$q_k = p_{11}(k). \quad (9)$$

The decision rule is

$$\hat{y}_k = \begin{cases} 1, & q_k \geq \tau_Q, \\ 0, & q_k < \tau_Q, \end{cases} \quad (10)$$

where τ_Q is selected exclusively on the training episodes.

5 Supervisory Safety Semantics

The discrete layer provides an execution interpretation for the continuous decision score. When q_k remains below the warning boundary, the marking remains in P_N . Persistent intermediate evidence activates P_W or P_D , while sustained high corroborated evidence activates P_E . The exact benchmark reported in this paper evaluates the binary Emergency decision because it allows a direct comparison with the baselines. The four-mode Petri structure remains the application architecture rather than an additional source of statistical evidence.

The principal safety invariant is

$$M(P_N) + M(P_W) + M(P_D) + M(P_E) = 1. \quad (11)$$

A transition firing conserves this total token count. Therefore, the supervisory state cannot contain two simultaneous mutually exclusive modes under the declared execution semantics.

A second requirement is decision traceability. Every emergency decision is associated with the pair (λ_A, λ_B) and therefore with the two sensor groups that produced it. This permits post-event diagnosis: a decision caused by strong corroboration is distinguishable from a decision caused by a single-channel excursion.

A third requirement is fail-conservative interpretation. The benchmark does not interpret a low QFLPN score as proof that the physical system is safe. It only means that the measured evidence did not cross the configured decision boundary. This distinction is essential in safety-critical systems because absence of an alarm is not equivalent to certification of safe operation.

6 Baseline Methods

Three decision mechanisms are evaluated. The first is a crisp guard

$$r_k = \max(\mu_T, \mu_V, \mu_P, \mu_I), \quad (12)$$

which represents a permissive event guard in which one sufficiently abnormal sensor can activate the alarm.

The second is fuzzy fusion

$$f_k = \frac{\lambda_A + \lambda_B}{2}. \quad (13)$$

This baseline preserves graded evidence but does not require multiplicative corroboration between the two evidence groups.

The third is the QFLPN score

$$q_k = \lambda_A \lambda_B, \quad (14)$$

which is exactly the $|11\rangle$ measurement probability produced by the declared two-qubit encoding.

All three thresholds are selected on the same 180 training episodes using the same F1 objective. No threshold is selected on the held-out test episodes. This avoids the common error of optimizing the decision boundary after observing the final results.

7 Experimental Design

The benchmark contains 300 episodes, each with 240 samples. The pseudorandom generator uses seed 20260908. The first 180 episodes form the training set and the final 120 episodes form the held-out test set. Each episode contains one fault event with a randomly generated onset between samples 70 and 150 and a duration between 40 and 90 samples.

The latent severity is zero before the event and then follows a deterministic rise-and-recovery profile. Gaussian measurement noise is added independently to the four sensor channels. Three fault families are used, with each family perturbing at least two modalities. The binary emergency ground truth is one when the latent severity reaches at least 0.55.

Table 1: Synthetic benchmark configuration.

Parameter	Value
Episodes	300
Samples per episode	240
Training episodes	180
Held-out test episodes	120
Sensor channels	4
Fault families	3
Fault onset	70–150 samples
Fault duration	40–90 samples
Emergency severity threshold	0.55
Random seed	20260908
Quantum qubits	2
Numerical quantum simulation	exact state vector

The evaluation uses F1 score, precision, recall, and false-positive rate. Robustness is evaluated by injecting a positive temperature bias into normal test samples only. Bias values are 0, 0.4, 0.8,

1.2, and 1.8 normalized units. The true physical label is not modified by the corruption. This construction therefore measures susceptibility to a false sensor signal rather than a change in the physical process.

8 Implementation and Reproducibility

The benchmark is implemented as a deterministic numerical reference experiment. Sensor generation, fuzzy membership, QFLPN encoding, measurement probability, threshold optimization, and evaluation are all executed from the same parameter specification. The exact row-level dataset is stored with the manuscript artifacts.

The quantum state is simulated classically. For the two-qubit system used here, the state vector contains four complex amplitudes. The simulation is therefore intended to validate semantics and decision behaviour, not to demonstrate a quantum-hardware speedup.

The reproducibility contract includes the random seed, episode count, sample count, sensor equations, fault construction, membership parameters, evidence-group coefficients, train-test split, threshold-search grid, and attack amplitudes. Any change to these quantities constitutes a new experiment rather than an alternative presentation of the same result.

9 Clean Test Results

Table 2 reports the held-out clean test results. The crisp guard produces the highest F1 score, demonstrating that a permissive single-channel rule can be highly effective when the test data are clean. QFLPN is slightly below this value but above the fuzzy-fusion baseline. Its precision is the highest of the three methods, while recall is lower than the crisp guard.

Table 2: Held-out clean test results. Thresholds are selected only on the training set.

Method	Threshold	F1	Precision	Recall	FPR
Crisp guard	0.880	0.9084	0.8938	0.9235	0.0230
Fuzzy fusion	0.570	0.8421	0.8169	0.8688	0.0408
QFLPN	0.315	0.8499	0.8751	0.8262	0.0247

The clean result is deliberately not presented as evidence that QFLPN universally outperforms conventional methods. The strongest baseline is better in this regime. The scientific value of the QFLPN construction appears when the observation model is stressed.

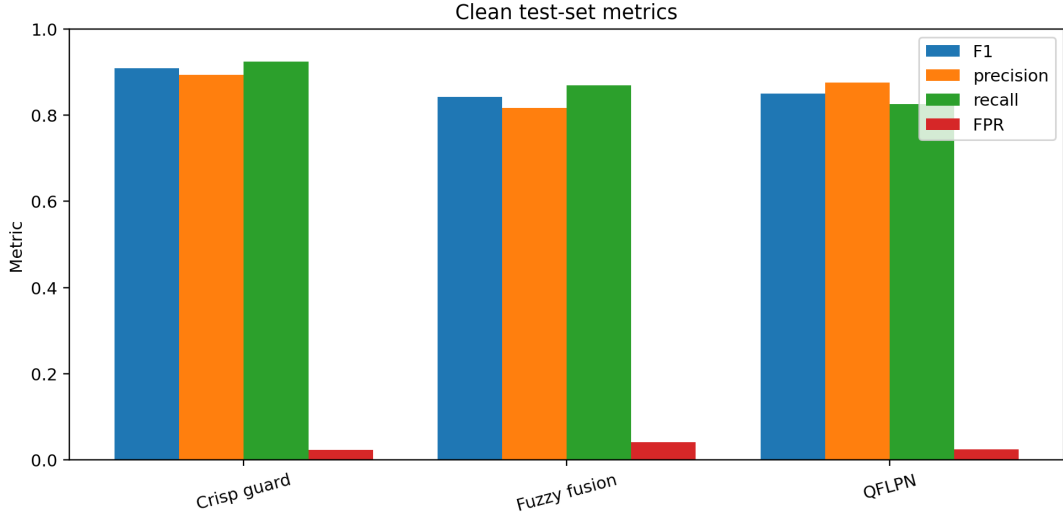


Figure 1: Clean held-out test metrics. The QFLPN rule provides higher precision than fuzzy fusion but lower recall than the crisp guard.

10 Robustness to Single-Sensor Corruption

The principal experiment applies a systematic temperature bias to normal test samples. The physical ground truth remains normal, so any alarm generated by the biased temperature is a false alarm. The experiment therefore tests whether the decision architecture can reject a single-channel observation that is inconsistent with the other sensing modalities.

The results are shown in Table 3. At zero bias, QFLPN has $F1=0.8499$ and a false-alarm rate of 0.0247. At a bias of 0.8, the QFLPN false-alarm rate is 0.0859, compared with 0.1039 for fuzzy fusion and 0.2870 for the crisp guard. At 1.2, the gap becomes substantial: QFLPN remains at 0.0859, while fuzzy fusion rises to 0.1443 and the crisp guard reaches 0.9933. At the strongest tested bias, 1.8, the crisp guard alarms on all normal corrupted samples, whereas QFLPN remains below 0.09 false-alarm rate.

Table 3: Robustness to a positive temperature bias applied only to normal test samples.

Bias	Crisp guard FPR	Fuzzy fusion FPR	QFLPN FPR
0.0	0.0230	0.0408	0.0247
0.4	0.0537	0.0990	0.0815
0.8	0.2870	0.1039	0.0859
1.2	0.9933	0.1443	0.0859
1.8	1.0000	0.2071	0.0859

The result is structurally interpretable. A temperature-only corruption strongly increases λ_A because temperature belongs to the thermal-current evidence group. The vibration-pressure group λ_B remains comparatively low. The product probability $p_{11} = \lambda_A \lambda_B$ therefore remains bounded by the uncorrupted group’s evidence. The architecture consequently behaves as a corroboration gate.

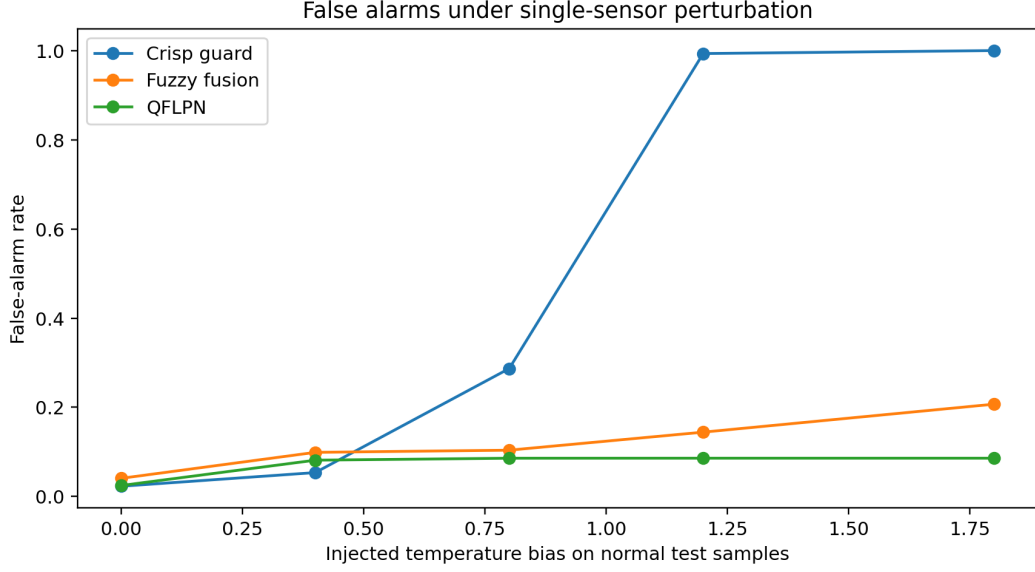


Figure 2: False-alarm rate under progressively stronger temperature bias on normal test samples.

11 F1 Robustness Curve

False-alarm behaviour is directly reflected in overall decision quality. At a temperature bias of 1.2, QFLPN reaches $F1=0.7390$, compared with 0.6796 for fuzzy fusion and 0.2773 for the crisp guard. At the strongest tested bias of 1.8, QFLPN remains at $F1=0.7390$, while fuzzy fusion decreases to 0.6084 and the crisp guard to 0.2760.

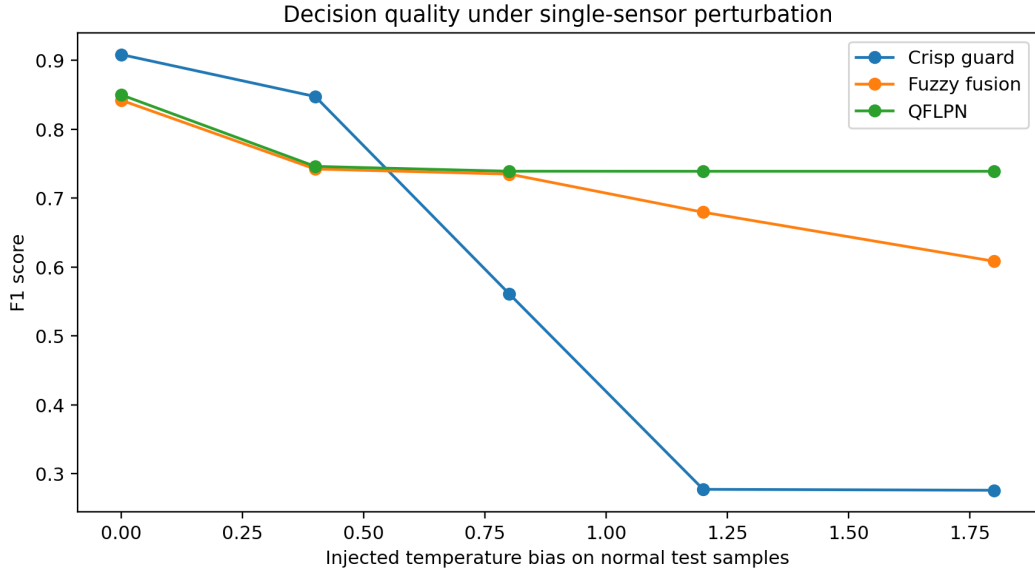


Figure 3: Held-out F1 score as a function of single-sensor temperature bias.

This result is more informative than a clean-data ranking because it exposes the operating regime in which the architectures behave differently. The crisp guard is highly effective when sensors are trustworthy, but it is intrinsically vulnerable to a sufficiently strong single-channel excursion. Fuzzy fusion is more stable, while the QFLPN product construction provides the strongest stability among the graded methods in the tested corruption range.

12 Fault-Family Analysis

The three fault families were designed so that no single modality uniquely identifies every event. The QFLPN rule therefore has an opportunity to benefit from cross-modal evidence while still retaining a measurable failure regime.

Table 4: F1 score by fault family on the held-out test set.

Fault family	Crisp guard	Fuzzy fusion	QFLPN
Thermal-pressure	0.8761	0.8710	0.8906
Mechanical	0.9223	0.8695	0.8883
Pressure-dominated	0.9266	0.7808	0.7742

QFLPN performs better than both baselines for the thermal-pressure family and below the crisp guard for the other two families. The pressure-dominated family exposes an important limitation: a multiplicative corroboration rule can suppress legitimate events when one evidence group reacts more weakly than the other. This is not a numerical defect; it is a design trade-off created intentionally by the safety-oriented corroboration mechanism.

13 Representative End-to-End Trace

Figure 4 shows a representative held-out episode. The latent severity rises during the fault event, while the QFLPN probability follows the joint activation of the two evidence groups. The decision threshold is crossed only when the corroborated evidence is sufficiently strong.

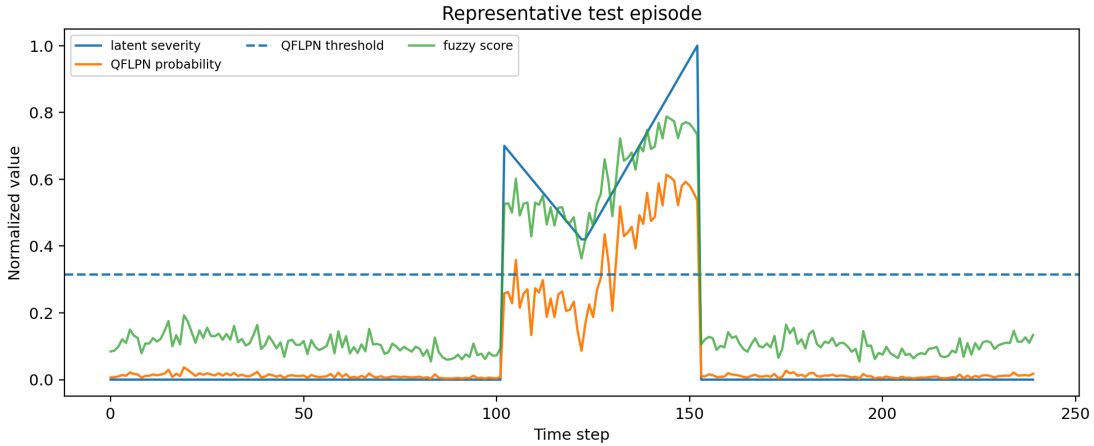


Figure 4: Representative held-out episode showing latent severity, QFLPN probability, fuzzy score, and the learned QFLPN threshold.

The figure should not be interpreted as a physical response curve. It is a controlled computational trace showing the complete transformation from noisy measurements to a final decision score.

14 Formal Safety Properties

The application has three mathematical properties that are independent of the statistical benchmark.

First, the quantum probability is bounded:

$$0 \leq p_{11} = \lambda_A \lambda_B \leq 1. \quad (15)$$

Second, the supervisory token invariant is preserved by every legal mode transition:

$$\sum_{p \in P} M(p) = 1. \quad (16)$$

Third, the corroboration score is monotone in each evidence group:

$$\frac{\partial p_{11}}{\partial \lambda_A} = \lambda_B \geq 0, \quad \frac{\partial p_{11}}{\partial \lambda_B} = \lambda_A \geq 0. \quad (17)$$

The last property means that stronger evidence in either group cannot reduce the QFLPN probability when the other group is fixed.

The same property explains the central robustness result. If one group remains bounded by $c < 1$, then regardless of the corrupted value of the other group,

$$p_{11} \leq c. \quad (18)$$

Thus, a threshold $\tau_Q > c$ cannot be crossed by a single corrupted group alone. This is a mathematical explanation of the measured false-alarm plateau.

15 Comparison with Alternative Assurance Strategies

The comparison is not intended to establish that QFLPN replaces formal verification, runtime verification, fuzzy control, or Petri-net analysis. Instead, the architectures address different layers of the assurance problem.

Classical Petri nets provide discrete structure and concurrency. Fuzzy fusion provides graded evidence aggregation. Runtime verification provides trace-level property checking. Timed-automata approaches can provide strong offline guarantees for specified timing models. The QFLPN application layer combines discrete supervisory structure, graded evidence, and an explicit operatorial state encoding.

Recent CPS runtime-verification research demonstrates that temporal logic can provide predictive anomaly properties for observed traces [12]. Formal verification plus runtime monitoring can similarly bridge model-level guarantees and executable systems [13]. These methods remain complementary. A mature critical-system architecture could use a QFLPN supervisor inside a broader assurance stack in which invariants, temporal properties, fail-safe states, and runtime monitors are independently checked.

16 Discussion

The principal scientific result is conditional. The QFLPN construction does not win the clean-data benchmark against the crisp guard. Instead, it provides a more stable decision boundary under the specific perturbation in which one sensor is systematically corrupted while the physical state remains normal.

This behaviour follows directly from the architecture. The crisp guard implements an OR-like decision and therefore reacts strongly to one abnormal sensor. Fuzzy averaging attenuates the effect but still permits one group to dominate. QFLPN uses the probability of a joint basis state, which implements an AND-like corroboration between the two evidence groups.

The result has an important engineering implication. Robustness should be measured against the failure modes of the observation layer, not only against clean test accuracy. A method that wins on clean F1 but fails catastrophically under a plausible sensor corruption may be inappropriate for a safety-monitoring role. Conversely, a method that is robust to one failure mode can lose sensitivity to legitimate events whose evidence is inherently asymmetric.

The pressure-dominated fault family illustrates this trade-off. The QFLPN score is lower for this family because one evidence group may remain weaker. The architecture therefore requires either calibrated group construction, additional sensor modalities, adaptive thresholds, or a mode-dependent decision policy if the application demands high sensitivity to asymmetric faults.

The application also clarifies what the quantum layer contributes. In this experiment, it provides a formally defined state representation and measurement interface. It does not provide a computational speedup, and the state is simulated classically. The scientific claim is therefore about the semantics and decision architecture, not about quantum hardware performance.

17 Limitations and Threats to Validity

The benchmark is synthetic. Its purpose is controlled mechanism validation, not industrial qualification. Real CPS data contain non-Gaussian noise, missing measurements, calibration drift, temporal correlation, communication delay, actuator saturation, and complex fault propagation.

The fault model contains three families and a single primary corruption mechanism. Other attacks or failures, including vibration spoofing, pressure bias, correlated multi-sensor corruption, sensor dropout, delayed measurements, and adversarially coordinated faults, are outside the present evidence.

The threshold is optimized for F1 on the training episodes. Other objectives may be more appropriate for safety-critical systems, such as minimizing missed emergencies subject to a hard false-alarm constraint. The present threshold-selection procedure therefore demonstrates reproducibility rather than prescribing a universal operational policy.

The quantum state is only two qubits and is simulated exactly. No claim about scalability, fault-tolerant quantum computing, quantum advantage, or physical-device noise is supported. The application should not be described as quantum hardware validation.

Finally, the Petri invariant is structural. It does not prove that the physical plant is safe. Safety certification would require system-specific hazard analysis, requirements traceability, verification evidence, validation data, and appropriate standards compliance.

18 Evidence Boundary and Scientific Contribution

The evidence supports four statements. First, the proposed end-to-end pipeline is mathematically well-defined for the declared sensor, fuzzy, Petri, and quantum layers. Second, the QFLPN probability can be interpreted as a corroboration score between two evidence groups under the specified encoding. Third, the measured held-out experiment demonstrates improved robustness to the tested single-sensor temperature corruption relative to the two graded or crisp baselines in the reported stress regime. Fourth, the complete benchmark can be reproduced from the declared seed and parameterization.

The evidence does not support universal superiority, industrial certification, quantum hardware advantage, asymptotic complexity improvement, or safety guarantees for arbitrary CPS. These claims require separate experiments and formal evidence.

The article’s distinct scientific contribution is therefore the application-level synthesis and controlled robustness experiment. The result is not obtained by reusing a runtime benchmark, a

sparse matrix benchmark, an open-system benchmark, or a tensor-network storage benchmark. The measured quantity here is safety-decision robustness under cross-modal sensor corruption.

19 Reproducibility and Data Availability

The row-level dataset is provided as `P10_safety_supervision_dataset.csv`. The robustness sweep is provided as `P10_attack_robustness.csv`, while the clean metrics are stored in `P10_clean_metrics.csv`. These artifacts contain the numerical values used for the tables and figures.

A complete reproduction requires Python-compatible numerical linear algebra, the stated random seed, the sensor-generation equations, the membership parameters, the train-test partition, and the threshold grid. The quantum calculation requires only a four-amplitude state-vector representation for the present two-qubit case.

The distinction between data and evidence is retained throughout. The dataset demonstrates the synthetic experiment; it does not constitute evidence that a physical industrial system has the same statistical distribution.

20 Conclusion

This paper developed an end-to-end QFLPN safety-supervision architecture that connects noisy sensor measurements, fuzzy evidence, discrete Petri-state supervision, quantum state encoding, and a final emergency decision.

On clean held-out data, the QFLPN method achieved $F1=0.8499$, below the crisp guard’s 0.9084 but above fuzzy fusion’s 0.8421 . The more important result emerged under single-sensor corruption. With a temperature bias of 1.2 , QFLPN maintained a false-alarm rate of 0.0859 , while fuzzy fusion reached 0.1443 and the crisp guard 0.9933 . At the strongest tested bias of 1.8 , QFLPN retained $F1=0.7390$ while the crisp guard fell to 0.2760 .

The experiment therefore supports a specific conclusion: cross-modal corroboration encoded through the declared QFLPN probability can make a safety decision substantially more resistant to single-sensor corruption in the tested synthetic CPS workload. The same mechanism can reduce sensitivity to legitimate asymmetric faults, as demonstrated by the pressure-dominated family.

The appropriate next scientific step is not to claim universal superiority, but to extend the experiment to real or public CPS datasets, multiple corruption mechanisms, timing uncertainty, missing data, adaptive thresholds, and formal temporal safety properties. Only after such evidence is obtained should stronger claims about deployment readiness or broad superiority be considered.

Author Contributions

Adrian Roman developed the QFLPN application formulation, mathematical model, experimental protocol, numerical implementation, analysis, and manuscript preparation.

Data and Code Availability

The manuscript artifacts include the generated dataset, robustness sweep, clean metrics, and figure data used for the reported results. The quantum layer is a classical exact simulation and is fully specified by the equations in the manuscript.

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